

JOEL KALLARACKAL

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EDUCATION

Indian Institute of Technology, Madras

Bachelors of Technology, Engineering Physics

CGPA - 8.79/10

Oct 2022 – Present

PUBLICATIONS

- Hanlin Shi, **Joel Kallarackal**, Karthik Karumanchi, Sylvain Pellegrini, Andrew Orekhov, Andy Vu, Bhaskar Vundurthy, and Howie Choset. "A Unified Compliant Locomotion Framework for Snake Robot Navigation in Complex External Pipe Networks." Submitted to IEEE Transactions on Automation Science and Engineering (T-ASE).

RESEARCH & ENGINEERING EXPERIENCE

Motion Planning and Control of Snake Robots

Jul 2025 – Jul 2026

Prof. Howie Choset, Robotics Institute, Carnegie Mellon University

Pittsburgh, PA

Kinodynamic Motion Planning for a Modular Underwater Snake Robot

- Developed a motion-planning framework integrating Interleaved Search and Trajectory Optimization (**INSAT**) with Graph of Convex Sets (**GCS**) to generate collision-free, dynamically feasible trajectories.
- Developed **dynamics model** incorporating joints, propeller thrust, gravity, buoyancy, and hydrodynamic drag.
- Developed an IMU-based attitude estimator using an **error-state Kalman filter(ESKF)** for orientation estimation.
- **Calibrated** underwater camera intrinsics and camera-IMU extrinsics and evaluated **visual SLAM** performance.
- Characterized the robot's **stable configuration manifold**, reducing the dimensionality of the planning problem.

Anchoring and Reaching Behavior for Confined Space Navigation

- Assembled, integrated and tested a **16-link modular snake robot** with 90° joint offsets between consecutive links.
- Represented robot with **Bézier curves**, reducing planning from 16 to 6 dimensions while preserving feasible motion
- Deployed a kinodynamic motion planner, demonstrating autonomous **multit-reaching behavior** on the snake robot.

Complex Pipe Network Navigation

- Simulated **contact-aware compliant control** using torque sensing to adapt body curvature to varying pipe diameter
- **Deployed** controller on the snake robot for pipe network navigation, **reducing contact torque** across diverse gaits.
- Simulated **windowed rolling-helix** gait for T-junction navigation, and **spiraling** gait for obstacle avoidance.

Game-Theoretic Multi-Agent Ergodic Search

Oct 2025 – Mar 2026

Prof. Howie Choset, Robotics Institute, Carnegie Mellon University

Pittsburgh, PA

- Formulated deceptive strategies for the agent as a linear program, achieving **higher payoff** than naive fictitious play
- Integrated ergodic search and deceptive strategies **maximizing coverage** while **minimizing information loss**
- Developed and deployed a **receding horizon ergodic controller** on four TurtleBots combining SMC & pure pursuit.

Autonomous Ground Vehicle Development

Mar 2023 – Nov 2024

Prof. Prabhu Rajagopal, Prof. Sathyan Subbiah, Center for Innovation, IIT Madras

Chennai, India

Autonomous Electric Golf Cart

- Developed and deployed a **ROS-based autonomy stack**, transforming an electric golf cart into a driverless shuttle.
- Implemented a **GPS waypoint generation and tracking system** for autonomous route planning and navigation.
- Fused depth data from stereo-camera with **segmented non-drivable region** from image for **costmap generation**.
- **Accelerated** costmap generation of non-drivable region obtained via semantic segmentation **300x** by parallelization

Intelligent Ground Vehicle Competition (IGVC), Michigan, USA

- **Led a 9-member team** in the design, development, and testing of an autonomous differential-drive ground vehicle.
- Designed a **robust navigation stack** for the robot as a distributed ROS system on a Raspberry Pi 4 and Jetson Orin
- Developed an incline and ramp detection algorithm by clustering 3D point-cloud surface normals using DBSCAN.
- Secured 6th place in the Design Challenge and 8th overall among 30+ international teams.

Multi-Agent Pursuit Evasion of Heterogeneous USVs

Jul 2026 – Present

Prof. Abhilash Sharma, Department of Ocean Engineering, IIT Madras

Chennai, India

- Developed a **USV dynamics simulation** based on Fossen's equations for multi-agent reinforcement learning.
- Developed a PettingZoo based **MARL environment** for multi-USV **pursuit & encirclement** of an adversarial vehicle.

Visuomotor Reinforcement Learning for Soft Tissue Manipulation

Dec 2023 – Apr 2024

Prof. Nirav Patel, Department of Engineering Design, IIT Madras

Chennai, India

- Developed a visuomotor reinforcement-learning agent for **grasping target locations** on tissues using **DVRK PSM**.
- Developed latent representations using convolutional autoencoders **reducing dimensionality** of observation space.
- **Simulated** soft tissue with the SOFA framework and generated **synthetic dataset** for offline reinforcement learning.

Gaze Prediction from Lensless Camera Images

Dec 2024 – Apr 2025

Prof. Kaushik Mitra, Department of Electrical Engineering, IIT Madras

Chennai, India

- Developed transformer based **gaze-estimation model** from lensless images, achieving a **1.17° mean angular error**
- Reconstructed eye images from lensless measurements using a fine-tuned **Dense Prediction Transformer (DPT)**.

Localization and Control for a Skid-Steering Mobile Robot

Jun 2024 – Aug 2024

Prof. Suresh Sundaram, Department of Aerospace Engineering, Indian Institute of Science, Bangalore

Bangalore, India

- Developed a kinematic model and velocity control interface for Botsync Copernicus, a skid-steering mobile robot.
- Calibrated and deployed **LiDAR-inertial SLAM**, including **LIO-SAM** and **FAST-LIO** using Livox Mid 360 LiDAR.

COURSE PROJECTS

Koopman Modeling of Contact Dynamics for Robot Control

Jul 2026 – Present

Course: Data-Driven Control, Prof. Pudur Viswanadha Reddy

- Studying Koopman operator-based modeling to globally linearize contact-rich robot dynamics across contact modes.
- Developing a linear MPC controller using Koopman models for robot locomotion under changing contact conditions.

Port Hamiltonian Modelling and Control of Snakeboard

Jul 2025 – Nov 2025

Course: Geometry of Locomotion, Prof. Howie Choset

- Derived **reduced momentum coordinates** spanning allowed momentum subspace of the non-holonomic system.
- **Formulated & simulated** a port-Hamiltonian model; constructed **bond graphs** to analyze the system's energy flow.
- Developed a **energy based controller** which transfers energy from the torso to the forward momentum over time.

Multi Agent Planning with Deadlock Mitigation

Jan 2025 – May 2025

Course: Motion Planning, Prof. Nirav Patel

- Developed an A* variant with time as an additional planning dimension, enabling **emergent waiting behavior**.
- Handled diverse multi-robot deadlock scenarios that conventional A* planning fails to resolve.

Deep Learning with Differential Privacy

Jul 2024 – Nov 2024

Course: Introduction to Machine Learning, Prof. Sheetal Kalyani

- Integrated (ϵ, δ) -differential privacy in deep learning models to protect data confidentiality in computer vision tasks.
- Conducted experiments on **MNIST** and **CIFAR-10** datasets, evaluating the impact of privacy-preserving techniques.
- Applied **Differentially Private SGD** with gradient perturbation, gradient clipping, and adaptive composition

TECHNICAL SKILLS

Programming: Python, C++ , MATLAB/Simulink **Software & Simulation:** MuJoCo, PyBullet, Gazebo, CARLA, Solidworks, LTSpice, Mathematica **Robotics & ML:** ROS, ROS 2, Pinocchio, CasADi, Crocoddyl, Drake, PyTorch, Stable-Baselines3, Ray/RLlib, Gymnasium, PettingZoo, OpenCV, Open3D **Other:** Git, Docker, Linux, LaTeX

SELECTED COURSEWORK

Math: Differential Geometry, Probability, Linear Algebra, Information Theory

Robotics & Control: Geometry of Locomotion, Motion Planning, Classical Dynamics, Linear Dynamical Systems, Synthesis of Control, Data Driven Control, Optimal Control, Computational Geometry

Learning: Machine Learning, Deep Learning for Computer Vision, Reinforcement Learning